

Requirements Table — Test Case 3: Rubric Authoring, Test-Suite Binding & Version Control

Problem Statement #02 — Automated Rubric Assignment Evaluator

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SECTION: 5C

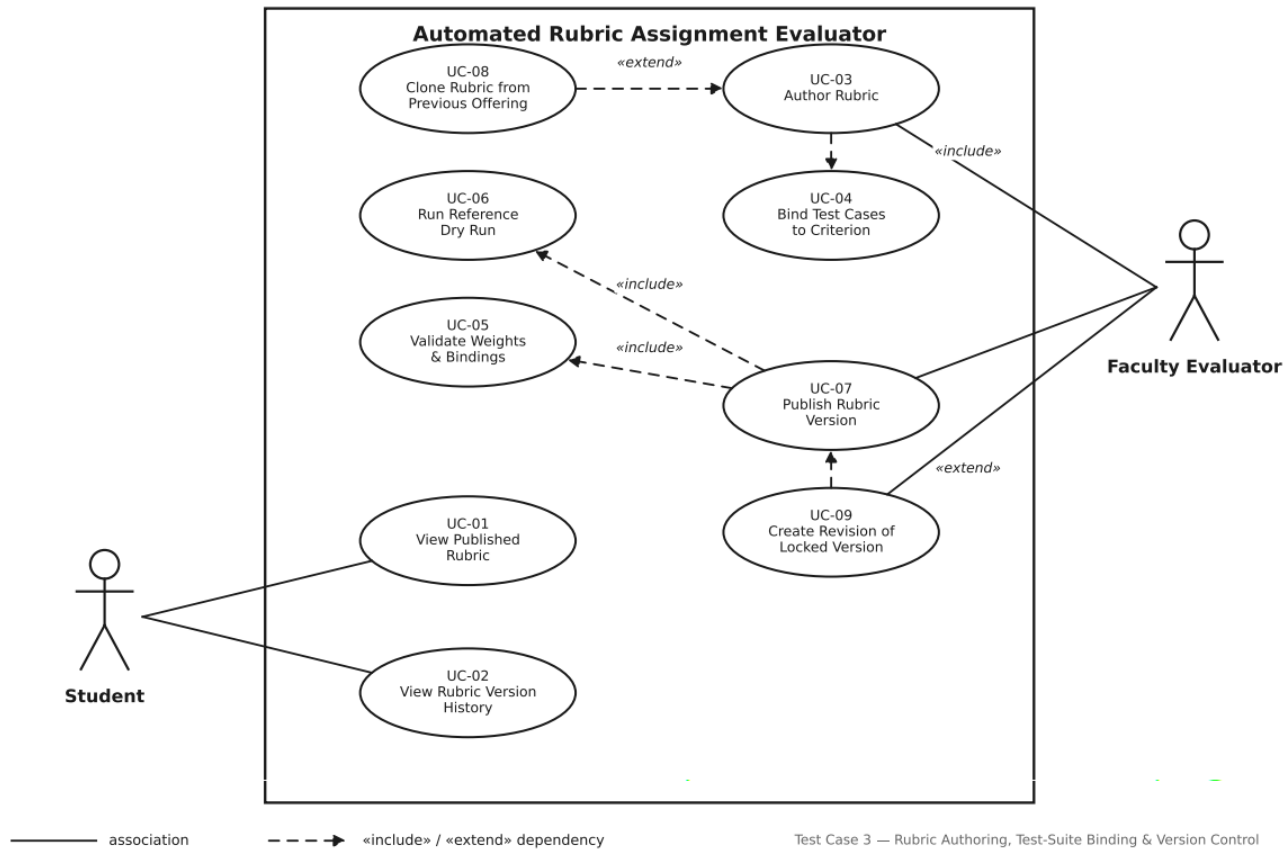
DEPARTMENT: AIML

SRN: PES1UG24AM154

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional — Rubric Authoring	The system shall allow a Faculty Evaluator to author a rubric as an ordered list of criteria, each with a title, descriptor, maximum marks, percentage weight and evaluation mode (automated, peer or manual).	High	Pass: a rubric with mixed evaluation modes is saved with criteria in the authored order and is retrievable unchanged. Fail: criterion order is lost, or a criterion is saved without a weight or maximum.	The rubric is the contract between the assignment and every score the system produces.
FR-002	Functional — Test-Suite Binding	The system shall allow each automated criterion to be bound to one or more test-case identifiers from the uploaded test suite, and shall reject publication while any automated criterion is unbound or any test identifier is unknown to the suite.	High	Pass: publication is blocked with the offending criterion or test ID named until every automated criterion is bound to existing tests. Fail: a rubric publishes with a dangling or unbound test reference.	An unbound criterion silently scores zero for the entire cohort — the most expensive failure mode.
FR-003	Functional — Weight & Partial-Credit Validation	The system shall reject any rubric whose criterion weights do not total 100%, and shall let the evaluator choose per criterion between proportional partial credit and all-or-nothing scoring.	High	Pass: a rubric totalling 100% saves; one totalling 97% or 103% is rejected with a field-level error; the chosen credit mode is applied at evaluation. Fail: an invalid total is persisted or applied to submissions.	Weight errors are invisible until grades are published, when they are costly to reverse.
FR-004	Functional — Dry Run Against Reference Solution	The system shall execute a published-candidate rubric and its bound test suite against the evaluator's reference solution and report the criterion-wise marks obtained before the rubric is released to students.	Medium	Pass: a dry run on a correct reference solution reports full marks on every automated criterion within 30 s; a deliberately broken reference reports the failing criteria. Fail: the dry run is skippable, or its result disagrees with a real evaluation of the same code.	The reference dry run is the only check that the rubric and the test suite actually agree.
FR-005	Functional — Versioning & Reuse	The system shall version every published rubric, lock a version once any submission has been evaluated against it, allow a new revision to be created from a locked version, and allow a rubric to be cloned from a previous course offering.	Medium	Pass: an edit to a locked version creates revision n+1 while evaluated submissions stay bound to version n; a cloned rubric carries criteria and bindings but not the published state. Fail: a locked version is mutated in place, or evaluated submissions change score retroactively.	Reproducing a published grade months later requires the exact rubric that produced it.

NFR-001	Data Integrity & Auditability	Every published rubric version shall be retained immutably with author, timestamp and content checksum, and shall remain retrievable for the institutional retention period; no delete operation shall remove a version bound to evaluated submissions.	High	Pass: a checksum comparison after a restore reproduces the original version byte for byte; a delete attempt on a bound version is refused. Fail: any published version is mutable, unrecoverable or deletable while in use.	Grade appeals and audits can arrive long after the offering closes.
NFR-002	Usability & Performance	The rubric editor shall load an existing rubric of up to 40 criteria within 2 s and return dry-run feedback within 30 s, with all validation errors reported in a single pass rather than one at a time.	Medium	Pass: a 40-criterion rubric renders within 2 s and an invalid rubric lists every error at once. Fail: errors are surfaced one per save attempt, or the editor stalls on large rubrics.	Rubric authoring happens under deadline pressure; slow iteration pushes staff back to spreadsheets.

USE CASE DIAGRAM



Use-Case Flow Specification — Test Case 3

Scope: Rubric Authoring, Test-Suite Binding & Version Control

Use Case ID	UC-07
Use Case Name	Publish Rubric Version
Primary Actor	Faculty Evaluator
Secondary Actors	Student (consumer of the published rubric); evaluation runner
Trigger	The Faculty Evaluator requests publication of a candidate rubric for an assignment
Related Requirements	FR-002, FR-003, FR-004, FR-005, NFR-001
Included Use Cases	UC-05 Validate Weights & Bindings, UC-06 Run Reference Dry Run

Preconditions

- The Faculty Evaluator is authenticated and owns the assignment.
- A candidate rubric exists with at least one criterion (UC-03).
- A test suite and a reference solution have been uploaded for the assignment.
- No submission has yet been evaluated against this candidate version.

Postconditions — success guarantee

- The rubric is stored as an immutable published version with number, author, timestamp and checksum.
- Every automated criterion is bound to existing test identifiers and weights total 100%.
- The published version becomes the active rubric for subsequent evaluations.
- The publication event is written to the audit log and the rubric becomes visible to students.

Postconditions — minimal guarantee

- No version is created; the candidate stays editable and the active rubric, if any, is unchanged.

Main Success Scenario

1. The Faculty Evaluator opens the candidate rubric and selects publish.
2. The system checks that criterion weights total 100% and that every criterion has a maximum and a mode — *«include» UC-05*.
3. The system verifies that every automated criterion is bound to test identifiers present in the uploaded suite.
4. The system executes the candidate rubric and its bound tests against the reference solution.
5. The system reports the criterion-wise dry-run marks and asks the evaluator to confirm.
6. The Faculty Evaluator reviews the dry-run result and confirms publication.
7. The system freezes the content, assigns the next version number and computes a content checksum.
8. The system marks the version active, writes the audit entry and locks the previous active version.
9. The system publishes the rubric to the assignment page for students.

Alternate Flow A1 — validation or dry-run failure

1. At step 2, 3 or 4 the weights do not total 100%, a criterion is unbound, a bound test ID is unknown, or the reference solution doesnot obtain full marks on an automated criterion.
2. The system abandons publication and creates no version.
3. The system reports every detected error in a single list, naming the criterion and the test identifier for each.
4. The Faculty Evaluator corrects the rubric or the test binding, or updates the reference solution.
5. The use case restarts at step 1; the previously active rubric remains in force until publication succeeds.